VIDEO WORLD MODELS SEE IN THE DARK: A CROSS-DOMAIN PRIOR FOR LOW-LIGHT VIDEO ENHANCEMENT

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ABSTRACT

Low-light video enhancers learn their prior from one paired dataset and degrade on others. We study whether a frozen video world model can supply a better prior. Measuring six frozen encoders on paired dark and well-lit clips, we find that illumination invariance is shared by self-supervised encoders, whereas V-JEPA's latent predictor is unique in forecasting future tokens of dark clips better than copying the last frame. We introduce Seer, which conditions a Restormer-style pixel branch on frozen V-JEPA tokens through residual cross-attention, a learnable cross-layer router and a counterfactual predictor that maps the dark clip's tokens towards those of its well-lit counterpart. On SDSD, Seer is the best of all evaluated methods in fidelity, perceptual quality and motion-compensated temporal consistency. A controlled study that changes only the tokens the pixel branch reads attributes the gain to the pretrained prior: $+ + 1.74$ dB on SDSD-indoor over the same network without V-JEPA, not reproduced by a randomly initialised encoder. On unseen datasets the prior does not help on its own. Our protocol corrects a frame-pairing error of earlier evaluations and reports the training-set membership of every released checkpoint; all numbers are generated from released per-sequence results.

1 INTRODUCTION

Low-light video enhancement (LLVE) is ill-posed: the same dark clip admits many plausible well-lit versions, and supervised models resolve the ambiguity with statistics learned from one paired dataset. They do well on held-out clips of that dataset and degrade on other sensors and scenes (Lin et al., 2024). A natural remedy is a prior that does not come from the paired data. Large frozen encoders are an obvious source, and among them video world models are attractive for video: V-JEPA (Bardes et al., 2024; Assran et al., 2025) is trained to predict the latent representation of masked spatio-temporal regions, so it models how scenes evolve rather than how pixels look.

We ask what such a prior actually buys for LLVE, and answer with controls that the conditioning literature rarely reports. First, we measure illumination invariance of six frozen encoders on paired low-light/well-lit clips (Sec. 3). V-JEPA is far more invariant than CLIP and SigLIP 2 but statistically tied with DINOv2/v3 and VideoMAE; what sets it apart is its predictor, which forecasts future latent tokens of dark clips far better than copying the last frame, while VideoMAE's pixel decoder does worse than copying. Second, we build **Seer**, which conditions a trainable Restormer-style pixel branch on frozen V-JEPA tokens through residual cross-attention, a learnable cross-layer router, and a counterfactual predictor that reuses V-JEPA's own predictor to map the tokens of the dark clip towards those of its well-lit counterpart (Sec. 4). Third, we evaluate on six levels from in-domain video to unseen image datasets, with a protocol that fixes a frame-pairing error of earlier evaluations, reports which test sequences each released checkpoint was trained on, and measures temporal consistency with a motion-compensated warping error (Sec. 5.1).

The controlled study (Sec. 5.3) keeps initialisation, schedule and seeds fixed and changes only the tokens the pixel branch reads. Pretrained V-JEPA tokens improve SDSD-indoor by $+1.74$ dB, lower LPIPS on both SDSD splits and make the output more temporally consistent, and a randomly

initialised encoder of the same size does not reproduce the gain. On unseen datasets, however, the same tokens do not help on their own.

Contributions. (i) A six-encoder study showing that illumination invariance is common to self-supervised encoders, whereas latent prediction under darkness is specific to the video world model. (ii) Seer, an LLVE model that reads a frozen V-JEPA through cross-attention, a cross-layer router and a counterfactual predictor, and is the best of all evaluated methods on the SDSD levels in fidelity, perceptual quality and temporal consistency. (iii) A controlled attribution of these gains to the pretrained prior, with random-encoder, encoder-swap and component controls, and an analysis of why they do not carry over to unseen datasets. (iv) An evaluation protocol with corrected frame pairing, per-checkpoint training-set membership, a retrained baseline on the same split and a motion-compensated temporal metric; all numbers are generated from released per-sequence results.

2 Related work

Low-light enhancement. Video methods exploit temporal redundancy through Retinex decomposition with cross-frame consistency (Xu et al., 2025), spatial-temporal compatible learning (Zhu et al., 2024), memory-controlled recurrence (Wang et al., 2025a), state-space models (Huang et al., 2025), event cameras (Kai et al., 2026) and diffusion (Lin et al., 2024; Ding et al., 2026); BVI-RLV (Lin et al., 2024) exposed their fragility across sensors. Image methods include Retinex transformers (Cai et al., 2023) and their Mamba and diffusion extensions (Weng et al., 2024; Wang et al., 2025b), HVI-CIDNet (Yan et al., 2025), DarkIR (Feijóo et al., 2025), URCSA (Dong et al., 2024), and unpaired or zero-reference methods (Guo et al., 2020; Ma et al., 2022; Wang et al., 2024b). Cross-dataset evaluation remains rare, and released checkpoints are often trained on sequences of common test splits (Sec. 5.1).

Pretrained representations as restoration priors. Frozen networks serve as perceptual judges (Johnson et al., 2016; Zhang et al., 2018; Ding et al., 2022; Wang et al., 2023) and as active conditioning: DA-CLIP (Luo et al., 2024) injects adapted CLIP embeddings into a restorer through cross-attention, pretrained features refine restoration networks (Xu et al., 2024), CLIP/DINO-style features guide diffusion restorers (Wang et al., 2024a), and illumination-invariant priors have been designed for low-light enhancement (Wang et al., 2024b). Seer follows this conditioning paradigm with a *video world model* (Bardes et al., 2024; Assran et al., 2025), which predicts masked spatio-temporal regions in latent space rather than pixels (Tong et al., 2022), and additionally reuses its predictor. We test the prior with controls the conditioning literature rarely reports: the same network without the tokens and with a randomly initialised encoder.

3 Which frozen encoder sees in the dark?

Illumination invariance. We feed six frozen ViT-L encoders the same 16-frame 512×512 low-light clip $\mathbf{X}$ and its well-lit ground truth $\mathbf{Y}$ from 15 SDSD-indoor pairs: CLIP (Radford et al., 2021), SigLIP 2 (Tschannen et al., 2025), DINOv2 (Oquab et al., 2024), DINOv3 (Siméoni et al., 2025), the pixel-reconstructing video autoencoder VideoMAE (Tong et al., 2022) and the latent-predictive video world model V-JEPA 2.1 (Assran et al., 2025). Image encoders are run per frame and averaged over time; all outputs are resampled to a common 16×16 grid (App. C). For the last-block tokens we measure the mean per-patch relative shift

$$\bar{\rho}_{\text{patch}}(\Phi;\mathbf{X},\mathbf{Y}) = \tfrac{1}{P}\textstyle\sum_p \|\Phi_p(\mathbf{X}) - \Phi_p(\mathbf{Y})\|_2 / \|\Phi_p(\mathbf{Y})\|_2. \tag{1}$$

Figure 1 shows two groups. The two image–text encoders drift most (SigLIP 2 0.603, CLIP 0.554); V-JEPA (0.431) is lower on every one of the 15 pairs (paired one-sided Wilcoxon $p{=}3.1\times10^{-5}$ against both). The self-supervised encoders form a single low-drift tier: DINOv2 0.462, DINOv3 0.434, V-JEPA 0.431 and VideoMAE 0.419; V-JEPA is statistically tied with DINOv3 ($p{=}0.47$) and is not more invariant than VideoMAE, which is lower on 10 of the 15 pairs. Illumination invariance therefore does not single out a video world model: it is shared by self-supervised encoders in general.

Latent prediction under darkness. What V-JEPA has and the other encoders lack is a predictor trained to forecast the latent tokens of unseen spatio-temporal regions. On 11 low-light clips (6

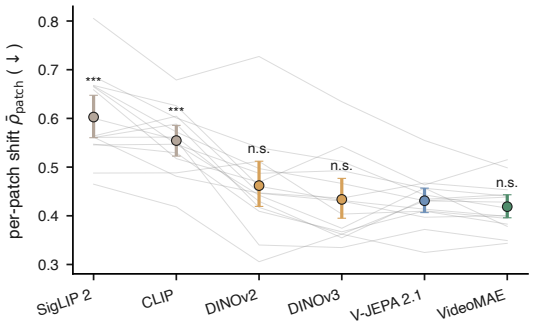


Figure 1: Single-frame illumination invariance of six frozen encoders on 15 SDSD-indoor pairs (lower is more invariant). Grey lines: individual pairs; markers: mean with bootstrap 95% CI. Stars: paired one-sided Wilcoxon test against V-JEPA (n.s.: $p>0.05$).

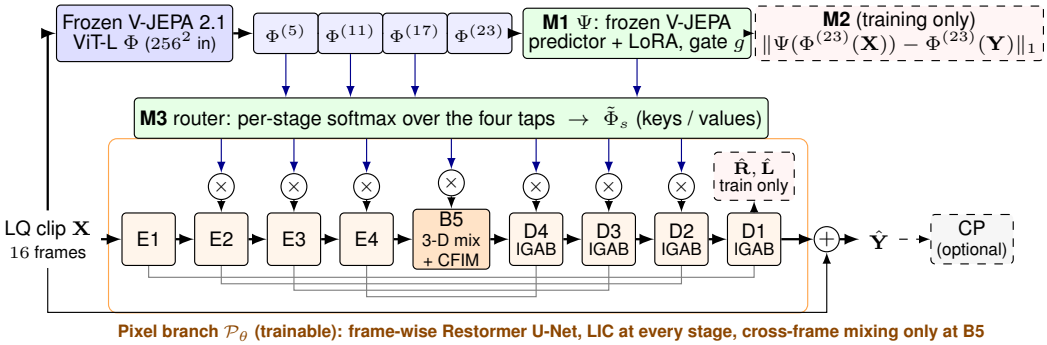


Figure 2: Seer. A frozen V-JEPA encoder turns the low-light clip into spatio-temporal tokens at four depths; a counterfactual predictor (M1) refines the deepest tap and is supervised to match the tokens of the well-lit clip (M2); a learnable router (M3) mixes the taps for each pixel-branch scale; the pixel branch reads them through zero-initialised residual cross-attention. Chroma preservation (CP) is an optional closed-form post-processor (Sec. 4.3).

BVI-RLV, 5 SDSD-outdoor), the frozen V-JEPA 2 predictor forecasts the tokens of held-out future tubelets with cosine 0.552 to the true tokens, against 0.458 for copying the last observed tubelet (paired Wilcoxon $p{=}5{\times}10^{-4}$, $d{=}1.46$, higher on $11/11$ clips). VideoMAE's pixel decoder, asked for the same future tubelets of the 15 SDSD-indoor low-light clips, is *worse* than copying (cosine to the true patches 0.406 vs. 0.558), and drops by 0.14 relative to well-lit input. The latent predictor is the component our method reuses (M1); Sec. 5.6 tests whether this difference matters once the encoder is plugged into a restoration network.

4 SEER

Notation. $\mathbf{X} \in [0,1]^{3\times T\times H\times W}$ is a low-light clip with $T{=}16$ frames, $\mathbf{Y}$ its well-lit counterpart and $\hat{\mathbf{Y}}$ the prediction. Φ is the frozen V-JEPA 2.1 ViT-L encoder (Assran et al., 2025) (304.7 M parameters, 24 blocks). It always receives the clip resized to $256{\times}256$, so every clip is represented by the same grid of $8{\times}16{\times}16{=}2048$ spatio-temporal tokens of width $d{=}1024$ (tubelet 2, patch 16), independently of the pixel resolution. We read the normalised outputs $\Phi^{(\ell)}$ of blocks $\ell \in \mathcal{L} = \{5, 11, 17, 23\}$ (0-indexed), the hierarchical taps used by V-JEPA 2.1 itself.

4.1 PIXEL BRANCH

The trainable pixel branch $\mathcal{P}_\theta$ is a frame-wise five-scale Restormer (Zamir et al., 2022) U-Net (widths 64–1024; $4, 8, 8, 12, 16$ encoder and $12, 8, 8, 4$ decoder blocks; pixel-(un)shuffle resampling) that predicts a residual, $\hat{\mathbf{Y}} = \mathbf{X} + \mathcal{P}_\theta(\mathbf{X}; \tilde{\Phi})$. Two low-level illumination pathways use only pixels:

local illumination cues (LIC; luma, channel maximum, 3×3 local mean, Gaussian low-pass luma and gradient magnitude of $\mathbf{X}$) are projected and added at every encoder and decoder stage, and Retinexformer's illumination estimator with illumination-guided attention blocks (IGAB, Cai et al., 2023) follows each decoder stage. Cross-frame interaction inside the pixel branch is limited to the $1/16$-resolution bottleneck (a residual 3-D temporal convolution followed by cross-frame attention, CFIM); all other temporal information reaches the pixel branch through V-JEPA's tokens.

Reading V-JEPA. At encoder and decoder stages $s \in \{2, \dots, 5\}$ (all but full resolution) the pixel features $\mathbf{F}_s$ of frame t attend to the 256 tokens of tubelet $\lfloor t/2 \rfloor$ of the routed feature $\tilde{\Phi}_s$,

$$\mathbf{F}_s \leftarrow \mathbf{F}_s + \mathbf{W}_o^s \,\mathrm{softmax}\Big(\frac{(\mathbf{F}_s\mathbf{W}_Q^s)(\tilde{\Phi}_s\mathbf{W}_K^s)^\top}{\sqrt{d_h}}\Big)\,\tilde{\Phi}_s\mathbf{W}_V^s, \tag{2}$$

with multi-head attention over the key axis and $\mathbf{W}_o^s$ initialised to zero, so that training starts from the plain pixel branch and V-JEPA acts as a read-only key/value memory.

4.2 Three V-JEPA-centric mechanisms

M1: counterfactual predictor. V-JEPA's pretrained predictor maps visible context tokens to the latent tokens of unseen content. We reuse it to map the tokens of the low-light clip to those the encoder *would* produce for the well-lit clip, which is what we mean by counterfactual. For the deepest tap $z{=}\Phi^{(23)}(\mathbf{X})$,

$$\Psi(z) = z + g \cdot \mathbf{W}_{\text{new}}\,\mathcal{P}_{\text{LoRA}}(z), \tag{3}$$

where $\mathcal{P}_{\text{LoRA}}$ is the frozen 12-block, 384-wide V-JEPA 2.1 predictor with rank-16 LoRA (Hu et al., 2022) on every attention `qkv` projection, run on all 2048 tokens without masking, $\mathbf{W}_{\text{new}}:\mathbb{R}^{384}\to\mathbb{R}^{1024}$ is a new linear head, and g is a scalar gate initialised to 0. M1 has 0.69 M trainable parameters on top of 22.3 M frozen predictor weights.

M2: counterfactual feature loss. The target that makes Ψ counterfactual is explicit: V-JEPA's own tokens of the ground truth,

$$\mathcal{L}_{\text{M2}} = \big\|\Psi\big(\Phi^{(23)}(\mathbf{X})\big) - \Phi^{(23)}(\mathbf{Y})\big\|_1, \tag{4}$$

averaged over tokens and channels. Because Φ is frozen, M2 trains only M1; it does not back-propagate into $\hat{\mathbf{Y}}$.

M3: cross-layer router. Each stage mixes the four taps with its own softmax weights, $\tilde{\Phi}_s = \sum_{\ell\in\mathcal{L}} \pi_s^{(\ell)}\,\Phi_\star^{(\ell)}$ with $\pi_s = \mathrm{softmax}(\boldsymbol{\lambda}_s)$, where $\Phi_\star^{(\ell)}{=}\Phi^{(\ell)}(\mathbf{X})$ for the three shallower taps and $\Phi_\star^{(23)}{=}\Psi(\Phi^{(23)}(\mathbf{X}))$. The logits start at 4 on the tap used by prior V-JEPA-conditioned variants of this architecture (stages 2–5 read blocks $5, 11, 17, 23$) and 0 elsewhere, i.e. 0.948 of the mass on that tap.

Objective. With auxiliary reflectance and illumination heads $(\hat{\mathbf{R}}, \hat{\mathbf{L}})$ on the last decoder features and the Charbonnier loss ρ ($\epsilon{=}10^{-3}$),

$$\mathcal{L} = 1.5\,\rho(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda_R\,\rho(\hat{\mathbf{R}}\odot\hat{\mathbf{L}}, \mathbf{Y}) + \lambda_S\,\mathrm{TV}(\hat{\mathbf{L}}) + \lambda_M\,\mathcal{L}_{\text{M2}}, \tag{5}$$

with $(\lambda_R, \lambda_S, \lambda_M){=}(0.15, 0.02, 0.05)$ and a spatial total-variation prior. Only $\hat{\mathbf{Y}}$ is used at inference.

4.3 Closed-form chroma preservation

On unseen domains the raw prediction can recover luminance well yet carry a colour cast. CP fuses $\hat{\mathbf{Y}}$ with a surrogate that keeps the input's chromaticity at the predicted luminance,

$$\hat{\mathbf{Y}}_{\text{cp}} = \mathbf{X}\odot\frac{Y(\hat{\mathbf{Y}})}{Y(\mathbf{X})+\epsilon}, \qquad \hat{\mathbf{Y}}_c^\star = \alpha_c\,\hat{\mathbf{Y}}_{\text{cp},c} + (1-\alpha_c)\,\hat{\mathbf{Y}}_c, \tag{6}$$

where $Y(\cdot)$ is BT.601 luma computed on the sRGB values, $\epsilon{=}10^{-3}$, c indexes RGB channels and the output is clipped to $[0,1]$. Writing $\varepsilon_1{=}\hat{\mathbf{Y}}_{\text{cp},c} - \mathbf{Y}_c$, $\varepsilon_2{=}\hat{\mathbf{Y}}_c - \mathbf{Y}_c$ and $m_{ij}{=}\mathbb{E}[\varepsilon_i\varepsilon_j]$ (uncentred, so biases are included), the MSE-optimal weight within this sum-to-one, intercept-free family is

$$\alpha_c^\star = \frac{m_{22}-m_{12}}{m_{11}-2m_{12}+m_{22}}, \tag{7}$$

which we clip to $[0, 1]$ so the fusion stays convex (and set to $\frac{1}{2}$ if the denominator vanishes). No distributional assumption is needed beyond second moments; this is a closed-form least-squares estimate, not a Bayesian posterior. $\alpha^\star$ must be estimated from paired data, and *where* it is estimated matters (Sec. 5.8). We report two calibrations: **CP-src** fits Eq. (7) on 15 fixed clips of the SDSD *training* split, so no target data is used; **CP-tgt** cross-fits it on the target split (two folds by scene, each scored with the $\alpha^\star$ fitted on the other fold), which corresponds to calibrating on a few paired clips of a new deployment domain.

4.4 Training

Seer is trained on the SDSD-indoor and SDSD-outdoor training sequences with AdamW (lr 2×10^{-5}, weight decay 10^{-4}), batch 1 with 8-step gradient accumulation, 256×256 crops of 16-frame clips, random flips, gradient clipping at 1 and bf16 autocast. The architecture was grown in warm-started stages on the same split (App. A); every controlled comparison in this paper starts from the same Stage-3 checkpoint and fine-tunes each arm with the same schedule and the same three seeds, so that arms differ only in the manipulated factor.

5 Experiments

5.1 Protocol

Six evaluation levels. We train on the SDSD-indoor and SDSD-outdoor training sequences (Wang et al., 2021) and evaluate at six levels of increasing distance: **L0** SDSD-indoor (15 held-out sequences), **L1** SDSD-outdoor (25), **L2** BVI-RLV (Lin et al., 2024) (16 clips of 8 scenes; different sensor and noise), and the image sets **L3** LOL-v1 (Wei et al., 2018) (15), **L4** LOLv2-Syn and **L5** LOLv2-Real (Yang et al., 2020) (100 each). Videos are scored on the centre 16 frames with a 512×512 centre crop; images at full resolution, tiled to 16 frames for V-JEPA with the middle output frame scored. Input and ground-truth frames are paired by sorted frame index: six SDSD-indoor sequences store their input and ground-truth frames with offset file names, and pairing by name, as in our earlier evaluation code, silently compares different time steps.

Metrics. Per frame: PSNR on uint8 RGB, LPIPS-alex (Zhang et al., 2018), DISTS (Ding et al., 2022) and CLIPIQA (Wang et al., 2023). Across frames: tLP (LPIPS between adjacent output frames), MABD, flicker, and a warping error WE, the masked mean absolute difference between output frame $t{+}1$ and output frame t warped with RAFT (Teed & Deng, 2020) flow estimated on the ground truth, with occlusions removed by a forward–backward check. tLP, MABD and flicker reward any output that changes little, including one that fails to follow motion; WE does not. Seer numbers are means over three fine-tuning seeds ($\pm$ one standard deviation); differences between arms are paired over sequences with a 95% bootstrap interval.

Baselines and training-set membership. We compare with Zero-DCE, SCI, HVI-CIDNet, DarkIR, URCSA, STCD and Retinexformer using released checkpoints, run with their official inference code under the protocol above. TCE-Net (Zhu et al., 2024) releases no weights and RetinexMCNet (Wang et al., 2025a) no code, so they are discussed but not evaluated. Released SDSD checkpoints are trained on their authors' splits, which overlap ours (App. B, Table 8): Retinexformer's SDSD-outdoor model saw 19 of our 25 L1 sequences during training. We therefore also report (i) Retinexformer retrained with its official code on exactly Seer's training split, and (ii) the six L0 sequences that no evaluated SDSD checkpoint was trained on ("clean-6").

5.2 Main results

In-domain. On the two SDSD levels Seer is the best of all evaluated models on PSNR and LPIPS (Table 1): 27.34 dB / .124 on L0 and 21.93 dB / .181 on L1, although the released Retinexformer and STCD checkpoints were trained on part of these test sequences. Retinexformer retrained on exactly our split reaches 25.67 dB / .229 on L0 (Table 2); our pixel branch without V-JEPA matches its PSNR with a much lower LPIPS (.132), and the V-JEPA tokens add the rest. On the six L0 sequences that no SDSD checkpoint has seen (App. B), the retrained Retinexformer has the higher PSNR (29.07 vs. 29.29 dB) while Seer keeps the lower LPIPS (.126 vs. .215).

Method	Train data	**L0** SDSD-in PSNR↑	**L0** SDSD-in LPIPS↓	**L1** SDSD-out PSNR↑	**L1** SDSD-out LPIPS↓	**L2** BVI-RLV PSNR↑	**L2** BVI-RLV LPIPS↓	**L3** LOL-v1 PSNR↑	**L3** LOL-v1 LPIPS↓	**L4** LOLv2-Syn PSNR↑	**L4** LOLv2-Syn LPIPS↓	**L5** LOLv2-Real PSNR↑	**L5** LOLv2-Real LPIPS↓
Zero-DCE	unsup.	18.19	.429	17.98	.357	17.37	.259	14.80	.335	17.71	<u>.168</u>	18.00	.312
SCI	unsup.	23.48	.390	17.84	.343	17.79	.282	14.78	.339	15.43	.233	17.30	.307
HVI-CIDNet	LOL	19.54	.405	18.51	.306	<u>18.92</u>	<u>.190</u>	**23.21**	**.100**	**25.13**	**.045**	<u>20.60</u>	<u>.127</u>
DarkIR	LOL-Blur	20.65	.279	18.75	.312	**20.46**	**.150**	17.09	.211	<u>18.77</u>	.191	19.92	.232
URCSA†	SDSD+LOL	24.19	.265	19.60	.258	18.38	.221	<u>21.88</u>	<u>.110</u>	16.08	.230	**24.84**	**.081**
STCD†	SDSD-in	23.48	.196	15.24	.355	15.98	.311	17.44	.398	15.09	.454	16.50	.459
Retinexformer†	SDSD	26.53	.181	**22.00**	.222	17.89	.218	17.23	.241	15.23	.303	18.81	.243
Retinexformer (retrained)	SDSD*	25.67	.229	19.96	.232	18.59	.195	17.05	.293	17.06	.265	17.50	.306
Pixel branch w/o V-JEPA	SDSD*	25.61±0.87	.132±.005	21.01±1.31	.212±.011	17.91±0.48	.243±.005	16.68±1.23	.323±.015	16.18±0.38	.324±.001	18.04±0.54	.320±.011
Seer	SDSD*	**27.34**±0.27	**.124**±.002	<u>21.93</u>±0.55	**.181**±.004	17.93±0.11	.251±.002	16.27±0.20	.336±.002	16.07±0.09	.350±.002	18.51±0.21	.343±.002
Seer + CP-src	SDSD*	<u>27.31</u>±0.26	<u>.126</u>±.002	21.86±0.50	<u>.184</u>±.003	17.97±0.12	.253±.003	16.28±0.20	.325±.006	16.08±0.09	.346±.004	18.49±0.22	.331±.004

Table 1: Six-level evaluation (PSNR in dB). †Released checkpoint trained on part of our L0/L1 test sequences (App. B). SDSD*: trained on Seer's SDSD training split only. Seer: mean ± s.d. over three seeds. CP-src uses $\alpha^\star$ fitted on SDSD training clips only. **Bold**/<u>underline</u>: best/second per column. DISTS and CLIPIQA in App. E.

Architecture	Training data	**L0** PSNR	**L0** LPIPS	**L1** PSNR	**L1** LPIPS	**L2** PSNR	**L2** LPIPS
Retinexformer	SDSD-in	24.33	.223	14.51	.383	17.14	.270
Retinexformer	SDSD-in+out	25.67	.229	19.96	.232	18.59	.195
Seer w/o V-JEPA	SDSD-in+out	25.61±0.87	.132±.005	21.01±1.31	.212±.011	17.91±0.48	.243±.005
Seer	SDSD-in+out	27.34±0.27	.124±.002	21.93±0.55	.181±.004	17.93±0.11	.251±.002

Table 2: Training-data control on Seer's split (raw outputs). Retinexformer is retrained from scratch with its official code on the SDSD-indoor training sequences only, or on indoor and outdoor. Seer rows: controlled runs (mean ± s.d. over seeds). No indoor-only Seer is shown: our only indoor-fine-tuned Seer starts from stages trained on both subsets and would overstate indoor-to-outdoor transfer.

Cross-dataset. On the four unseen datasets Seer is not competitive without calibration. On BVI-RLV it reaches 17.93 dB, behind DarkIR (20.46) and the retrained Retinexformer (18.59); on the LOL image sets it trails every model trained on LOL. The required brightening differs strongly between datasets (median ground-truth-to-input luma ratio 2.6–7.2× across L2–L5 vs. 3.5–4.5× on SDSD), so a model trained on one dataset cannot know the target exposure; Sec. 5.4 separates this factor from restoration quality.

5.3 Does the V-JEPA prior matter? A controlled study

Table 3 isolates the prior. All arms start from the same Stage-3 checkpoint and are fine-tuned with the same schedule and seeds; they differ only in the tokens read by the pixel branch. Pretrained V-JEPA tokens improve SDSD-indoor by $+1.74$ dB (95% CI $[+0.82, +2.68]$) and lower LPIPS on both SDSD levels, and they make the output more temporally consistent (tLP and motion-compensated WE, Table 3). A randomly initialised V-JEPA of the same size is worse than no V-JEPA in-domain, so the gain comes from pretraining rather than from the extra attention pathway. Across datasets the picture reverses: on BVI-RLV the pretrained tokens change PSNR by $+0.03$ dB and *increase* LPIPS by $+0.008$, the random-init encoder does better than the pretrained one on L2, and on L3–L5 the pretrained tokens raise LPIPS as well (Table 1). The pretrained video world model is thus a strong prior for the domain it is fine-tuned on, and we find no evidence that it transfers to unseen sensors on its own. One caveat applies to all arms: the pixel branch was pre-trained with V-JEPA tokens up to Stage 3, which may disadvantage the arms without them (App. A).

5.4 Where does the cross-domain gap come from?

Exposure. GT-mean scores (Table 4) rescale each output frame to the mean luma of its ground truth, removing the dataset-specific exposure level that no source-trained model can infer. **Few-shot adaptation.** Each model is fine-tuned for 200 steps on four scenes of the BVI-RLV *training* split (`S01`, `S09`, `S17`, `S25`; both frame rates) and evaluated on the unchanged BVI-RLV test split and on L0 (Table 5).

V-JEPA tokens	seeds	L0 PSNR	L0 LPIPS	L1 PSNR	L1 LPIPS	L2 PSNR	L2 LPIPS	L0 temporal tLP	L0 temporal WE
No V-JEPA (zero tokens)	2	25.61±0.87	.132±.005	21.01±1.31	.212±.011	17.91±0.48	.243±.005	.054±.003	.0107±.0003
Random-init V-JEPA	2	24.91±0.22	.136±.002	19.52±0.14	.241±.011	18.64±0.25	.220±.006	.054±.000	.0105±.0001
Pretrained V-JEPA (Seer)	3	27.34±0.27	.124±.002	21.93±0.55	.181±.004	17.93±0.11	.251±.002	.047±.001	.0095±.0002
Δ pretrained − none		**+1.74**	**-0.008**	+0.93	**-0.031**	+0.03	**+0.008**	**-0.007**	**-0.0012**
Δ pretrained − random		**+2.43**	**-0.013**	+2.41	**-0.060**	**-0.70**	**+0.031**	**-0.007**	**-0.0010**

Table 3: Same Stage-3 initialisation, schedule and seeds; only the tokens fed to the pixel branch change. Zero tokens remove V-JEPA; random-init V-JEPA keeps the architecture but not the pretraining. Bold Δ: 95% bootstrap interval over sequences excludes 0.

Method (GT-mean)	L0 PSNR	L0 LPIPS	L1 PSNR	L1 LPIPS	L2 PSNR	L2 LPIPS	L3 PSNR	L3 LPIPS	L4 PSNR	L4 LPIPS	L5 PSNR	L5 LPIPS
Zero-DCE	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
SCI	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
HVI-CIDNet	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
DarkIR	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
URCSA†	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
STCD†	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Retinexformer†	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Retinexformer (retrained)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Pixel branch w/o V-JEPA	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Seer	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Δ Seer − w/o V-JEPA	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 4: GT-mean diagnostic: every output frame is rescaled to the mean luma of its ground truth before scoring, for all methods. This uses ground truth and is not a deployable result; it isolates errors other than global exposure.

5.5 Component ablations

Table 6 changes one component at a time. Opening the M1 gate (g_0=0.5), i.e. actually using the counterfactual predictor's delta, lowers PSNR on L0, L1 and L2: with the M2 target alone, the refined tokens are less useful to the pixel branch than the raw ones, which is consistent with the gate staying at $|g|<10^{-3}$ when it is free. The pixel-domain illumination cues (LIC) matter most off the training scenes, the bottleneck temporal mixing improves tLP and L2, and the Retinex auxiliary losses and IGAB have no consistent effect.

5.6 Encoder choice

Replacing V-JEPA by other frozen ViT-L encoders inside the same architecture (Table 10, App. D) gives the same picture as the controlled study: V-JEPA yields the best L0 PSNR and LPIPS and the best L1 LPIPS (CLIP has a slightly higher L1 PSNR), whereas on L2 all six encoders lie within 0.9 dB and V-JEPA is in the middle. Invariance does not predict the in-domain order either: VideoMAE, the most invariant encoder, is fourth on L0 PSNR and second-to-last on L1. The swap favours V-JEPA, since each swapped variant starts from a pixel branch adapted to V-JEPA tokens (App. D).

5.7 Temporal consistency

Seer has the lowest tLP of all methods on the three video levels and the lowest motion-compensated warping error on L0 (Table 7); on L1 and L2 its WE is within 0.001 of the best. Since WE penalises outputs that fail to follow the ground-truth motion, the low tLP is not an artefact of a static output. Among the baselines only the two-frame video model STCD is close on L0. The V-JEPA tokens contribute measurably on L0 (Table 3), much less on L1 and L2.

Model	steps	**L2** BVI-RLV test PSNR	LPIPS	tLP	**L0** (source) PSNR	LPIPS
w/o V-JEPA	0	17.91±0.48	.243±.005	.073±.000	25.61±0.87	.132±.005
	100	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
	200	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Seer	0	17.88±0.11	.252±.002	.072±.000	27.50±0.10	.123±.003
	100	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
	200	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Δ Seer − w/o	0	-0.02	**+0.009**	**-0.001**	**+1.89**	**-0.008**
Δ Seer − w/o	100	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Δ Seer − w/o	200	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 5: Few-shot adaptation: fine-tuning on four BVI-RLV training scenes (no test scene), starting from each arm's SDSD model; step 0 is zero-shot. Mean ± s.d. over two source seeds; bold Δ: 95% bootstrap interval over test clips excludes 0.

Variant (750 it.)	**L0** PSNR	LPIPS	**L1** PSNR	LPIPS	**L2** PSNR	LPIPS	**L0** tLP
Seer, full (3 seeds)	27.11±0.21	.124±.001	22.85±0.18	.168±.003	18.02±0.15	.246±.003	.048±.001
M1 gate opened (g_0=0.5)	**-0.24**	-0.001	**-0.38**	-0.000	-0.19	-0.002	-0.000
M3 soft router init (c=1)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
w/o Retinex aux. losses (λ_R=λ_S=0)	+0.16	-0.000	+0.18	**-0.004**	-0.33	**+0.004**	**+0.002**
w/o IGAB	-0.02	-0.000	**+0.67**	-0.003	+0.39	+0.002	+0.000
w/o LIC	**+0.35**	**-0.003**	**-1.30**	+0.000	**-0.32**	+0.003	**+0.002**
w/o temporal mixer + CFIM	+0.15	+0.001	-0.26	-0.003	**-0.22**	**+0.002**	**+0.006**
+ CP-src post-processing	-0.01	+0.001	-0.03	+0.002	+0.03	+0.001	+0.003

Table 6: One component changed at a time, each fine-tuned 750 iterations from the same Stage-3 checkpoint and compared with the full model at the same iteration. Δ relative to the full model; bold: 95% bootstrap interval excludes 0.

5.8 Chroma preservation

Calibrated on source data (CP-src), the closed-form coefficient is close to zero on the green and blue channels and CP leaves Seer's output essentially unchanged on every split (Table 1, App. F). Calibrated on a few paired target clips (CP-tgt) it adds +0.85 dB on BVI-RLV and +0.69 dB on LOLv2-Syn. With the same protocol, no baseline gains more than +0.34 dB on BVI-RLV, while our pixel branch without V-JEPA gains +0.63 dB (Table 13); CP-src changes PSNR by less than 0.3 dB for every method except where its coefficient is transferred to a different checkpoint (App. F). The cross-domain errors of our architecture thus contain a colour component that a target-calibrated CP can remove, independently of the V-JEPA tokens; without target data CP does not help.

6 Conclusion and limitations

A frozen video world model is a useful prior for low-light video enhancement in the domain the restorer is fine-tuned on: with everything else fixed, pretrained V-JEPA tokens improve fidelity, perceptual quality and temporal consistency, and neither a random encoder of the same size nor other frozen encoders match them. The prior does not, by itself, make the restorer robust to unseen sensors and scenes. Two components we designed for the prior did not pay off: the counterfactual predictor stays at identity when free and hurts when forced open, and closed-form chroma preservation only helps when calibrated on target data.

Limitations. All controlled arms share a pixel branch pre-trained with V-JEPA tokens, so a from-scratch comparison may change the size of the in-domain effect. Seer is large (370 M trainable and 305 M frozen parameters) and slow compared with image enhancers. Our model code is a re-implementation from the checkpoint layout; all controlled comparisons use the same code, but its absolute numbers differ from the archived reference checkpoint, whose encoder-swap runs we report separately. Results use three seeds for the main arms and one or two for ablations.

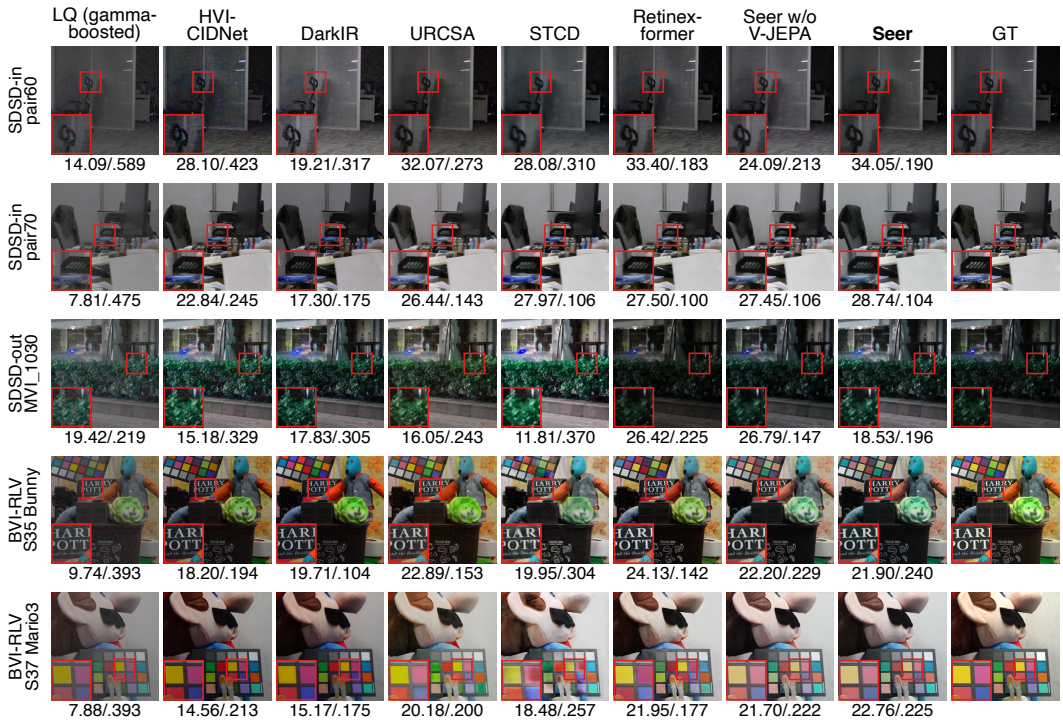


Figure 3: Frame 8 of five test clips: L0 `pair60` and `pair70` (not in the training set of any SDSD checkpoint shown), L1 `MVI_1030` (held out by Seer and by Retinexformer's release) and two BVI-RLV scenes. Red box: 2× zoom (inset, bottom left). Numbers: PSNR / LPIPS of the frame. The LQ column is gamma-boosted for display only; its scores are on the original input. Seer and Seer w/o V-JEPA: seed 0 of the controlled study (Sec. 5.3).

	L0		L1		L2	
Method	tLP	WE	tLP	WE	tLP	WE
GT	.030	.0065	.037	.0095	.069	.0057
HVI-CIDNet	.180	.0406	.133	.0331	.095	.0159
DarkIR	.132	.0183	.112	.0200	.081	**.0088**
URCSA†	.118	.0169	.094	.0172	.084	.0101
STCD†	.051	.0117	.067	.0166	.073	.0115
Retinexformer†	.095	.0148	.052	.0140	.079	.0111
Retinexformer (retrained)	.090	.0165	.081	.0201	.083	.0117
Pixel branch w/o V-JEPA	.054±.003	.0107±.0003	.051±.004	**.0123±.0011**	.073±.000	.0090±.0003
Seer	**.047±.001**	**.0095±.0002**	**.050±.002**	.0131±.0002	**.072±.001**	.0089±.0004
Seer + CP-src	.052±.004	.0099±.0004	.054±.002	.0136±.0004	.074±.002	.0094±.0007

Table 7: Temporal consistency on the 16-frame clips (↓). tLP: LPIPS between adjacent output frames; WE: warping error with RAFT flow from the ground truth and an occlusion mask. MABD, flicker and the LQ row are in App. G.

REPRODUCIBILITY STATEMENT

All numbers in this paper are generated by a script from per-sequence evaluation files (`tools/make_tables.py`); the training, evaluation and ablation code, the fixed split lists, the CP calibration clips and the per-sequence results will be released. Section 4 and App. A list every architectural and optimisation hyper-parameter; App. B lists the training-set membership of every evaluated checkpoint.

ETHICS STATEMENT

The work uses public research datasets (SDSD, BVI-RLV, LOL) and released model weights under their licences. Low-light enhancement can be used in surveillance; we do not target identification tasks.

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A Implementation details

V-JEPA input. The checkpoint is the released V-JEPA 2.1 ViT-L distilled from ViT-G at 384 px. Clips are bilinearly resized to 256×256 (so a 512×512 evaluation crop is downsampled by 2 for V-JEPA only), normalised with ImageNet statistics and encoded with RoPE, tubelet 2 and patch 16. Clips with fewer than 16 frames are repeat-padded. Taps are the outputs of blocks $5, 11, 17, 23$, each after its own hierarchical LayerNorm.

Pixel branch. Widths $(64, 128, 256, 512, 1024)$, attention heads $(4, 8, 8, 16, 16)$, GDFN expansion 2.66, bias-free LayerNorm. Each encoder and decoder stage is followed, in this order, by V-JEPA cross-attention (stages 2–5), LIC injection and, in the decoder, an IGAB with two blocks of illumination-guided channel attention (head width 64). The illumination estimator maps $[\mathbf{X}, \mathrm{mean}_c\,\mathbf{X}]$ through a 1×1 and a 5×5 convolution to 64 channels; a 1×1 convolution per decoder stage projects them to the stage width. The residual head is a zero-initialised 3×3 convolution; the reflectance and illumination heads are conv–SiLU–conv–sigmoid stacks with 3 and 1 output channels. The pixel branch has 369.8 M parameters; together with M1 the trainable total is 370.4 M, and the frozen V-JEPA encoder adds 304.7 M.

Cross-attention. Queries come from bias-free-normalised pixel features through a 1×1 convolution, keys and values from the routed tokens through one linear layer; attention is PyTorch scaled-dot-product attention with the stage's number of heads. The 8 V-JEPA tubelets are repeated twice along time to match the 16 frames.

Training stages. The Stage-3 checkpoint from which all controlled runs start was obtained by warm-started stages on the same SDSD training split with the optimiser of Sec. 4.4: an earlier V-JEPA-conditioned pixel branch with IGAB (`v34a`) was fine-tuned with M2 (Stage 1, 8k iterations), with the Retinex auxiliary heads (Stage 2, 10k), and with M1 (Stage 3, 8k). Stage 4 adds the M3 router; in this paper Stage 4 is the fine-tuning phase whose variants we compare. Because the pixel branch of every arm was pre-trained with V-JEPA tokens up to Stage 3, the no-V-JEPA arm of Sec. 5.3 measures the contribution of the tokens during the final fine-tuning and at inference; a from-scratch comparison is left for future work.

Controlled runs. Each arm is fine-tuned from Stage 3 for 1500 optimiser steps (ablations: 750) with the Stage-4 recipe, seeds $0, 1, 2$; audits during training are logged but not used for checkpoint selection, and the final checkpoint is evaluated. The code used here is a re-implementation of the model from its checkpoint layout (every tensor of the checkpoint loads with matching shape); all arms, including the reference, use this same code.

B Training-set membership of released checkpoints

For each released checkpoint we reconstructed its training sequences from the official option files and data loaders: the `testing_dir` lists of Retinexformer's SDSD indoor and outdoor option files (its loader excludes a folder when its name, or its name without `_2`, is listed), STCD's SDSD-indoor training options and URCSA's training script (both exclude by exact name only; URCSA releases no outdoor split). Retinexformer's SDSD release does not contain the six clean-6 sequences at all. Zero-DCE, SCI, HVI-CIDNet and DarkIR are trained without SDSD. SDSD records most scenes twice (`pairX`, `pairX_2`; the second pass is a different camera path through the same room), so exact-name exclusion can leave a test scene's second pass in the training set; we count only exact sequence overlap in Table 8. Seer's own split is the deterministic 80/20 split of sorted sequence names; it contains no L0 or L1 test sequence, but the second pass of two test scenes (`pair5`, `MVI_1007`) is in Seer's training set.

C Encoder invariance and predictive coherence

Tokenisation. V-JEPA and VideoMAE consume the 16-frame clip natively; their token grids are mean-pooled over time and average-pooled to 16×16. Image encoders are run on every frame at 512×512 and their per-frame token maps averaged over the 16 frames; CLIP and SigLIP 2 (patch 32)

Released checkpoint	L0 (15)	STCD-12	L1 (25)
Retinexformer†	7	5	19
URCSA†	0–9	0	unknown
STCD†	0–9	0	0

Table 8: Number of our test sequences contained in the training set of each released SDSD checkpoint, determined from the official split files (ranges: the release does not state whether second-pass `_2` sequences were included; unknown: no split file released). Seer's training split contains none of L0/L1. Details and file-level provenance in App. B.

Method	PSNR↑	LPIPS↓	DISTS↓	tLP↓	WE↓
Zero-DCE	20.28	.432	.236	.174	.0249
SCI	25.06	.399	.210	.153	.0244
HVI-CIDNet	22.30	.410	.235	.212	.0431
DarkIR	21.36	.289	.192	.153	.0193
URCSA†	26.44	.251	.182	.135	.0157
STCD†	24.15	.213	.177	.069	.0114
Retinexformer†	27.44	.182	.160	.111	.0147
Retinexformer (retrained)	29.07	.215	.172	.112	.0151
Pixel branch w/o V-JEPA	26.92±0.57	.137±.002	.139±.002	.074±.005	.0110±.0006
Seer	**29.29±0.16**	**.126±.004**	**.139±.001**	**.066±.002**	**.0097±.0003**
Seer + CP-src	29.26±0.18	.130±.004	.142±.002	.073±.006	.0102±.0005

Table 9: L0 restricted to the six sequences (`pair50/52/58/60/68/70`) that none of the evaluated SDSD checkpoints was trained on.

give 16×16 directly, DINOv2 (patch 14) and DINOv3 (patch 16) are run with interpolated position encodings and resampled in feature space. Averaging per-frame shifts instead of per-frame features raises DINOv3's shift from 0.434 to 0.534, so the aggregation choice favours the image encoders.

Predictive coherence. The predictor test uses the non-distilled V-JEPA 2 ViT-L checkpoint, whose predictor outputs the encoder's own 1024-d space (the distilled checkpoint used in Seer predicts a separate teacher space). For each clip (16 frames, 8 tubelets) the tokens of tubelets 4–7 are predicted from all earlier tubelets with the predictor's native masking; the naive baseline copies tubelet $t-1$. The VideoMAE analogue masks the same future tubelets and scores its decoder output against the normalised ground-truth pixel patches. The two scores live in different spaces and are not compared with each other, only with their own copy baselines.

D Encoder swap

	Invariance	L0		L1		L2	
Frozen encoder	$\bar{\rho}_{\text{patch}}$ ↓	PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS
V-JEPA 2.1 (video, latent pred.)	.431	26.66	.122	21.06	.168	16.66	.292
VideoMAE (video, pixel recon.)	.419	25.76	.135	19.11	.193	16.75	.294
DINOv3 (image)	.434	24.51	.155	20.20	.205	16.30	.314
DINOv2 (image)	.462	25.77	.136	19.38	.197	16.91	.278
CLIP (image-text)	.554	26.14	.131	21.39	.180	16.68	.308
SigLIP 2 (image-text)	.603	24.86	.146	18.99	.222	17.17	.304

Table 10: Frozen encoder swapped inside the full architecture. Invariance from Sec. 3. Swapped encoders were fine-tuned 1500 iterations from the V-JEPA Stage-3 checkpoint, the V-JEPA reference 6000; this favours V-JEPA (App. D). Raw outputs, no CP.

The swapped encoders replace V-JEPA inside the full architecture (all have width 1024, so the cross-attention projections are reused), M2 is computed in the swapped encoder's space, and each variant is fine-tuned from the V-JEPA Stage-3 checkpoint for 1500 iterations. The pixel branch therefore starts adapted to V-JEPA tokens, and the V-JEPA reference was fine-tuned for 6000 iterations; both choices favour V-JEPA. These runs, unlike the controlled study, used the original training code of the reference checkpoint.

E ADDITIONAL METRICS

	L0		L1		L2		L3		L4		L5	
Method	DISTS↓	CLIPIQA↑	DISTS↓	CLIPIQA↑	DISTS↓	CLIPIQA↑	DISTS↓	CLIPIQA↑	DISTS↓	CLIPIQA↑	DISTS↓	CLIPIQA↑
Zero-DCE	.235	.163	.186	.185	.143	.321	.216	.611	.129	**.513**	.204	.578
SCI	.209	.172	.176	.194	.145	.321	.210	**.621**	.161	.513	.202	**.592**
HVI-CIDNet	.233	.143	.170	.155	.110	.332	**.086**	.469	**.052**	.513	.099	.530
DarkIR	.192	**.419**	.191	**.391**	**.106**	**.444**	.166	.409	.162	.481	.183	.386
URCSA†	.195	.168	.151	.144	.133	.312	.093	.434	.168	.434	**.083**	.415
STCD†	.177	.263	.185	.161	.189	.265	.256	.217	.281	.239	.284	.194
Retinexformer†	.168	.248	.137	.262	.139	.309	.178	.386	.220	.320	.189	.317
Retinexformer (retrained)	.184	.216	.137	.195	.122	.345	.207	.303	.194	.403	.222	.257
Pixel branch w/o V-JEPA	.142±.000	.188±.001	.133±.004	.173±.001	.153±.008	.308±.010	.215±.010	.295±.005	.228±.002	.380±.025	.224±.007	.284±.002
Seer	**.139±.001**	.183±.003	**.113±.003**	.188±.002	.150±.003	.294±.007	.223±.003	.273±.003	.236±.002	.357±.008	.223±.002	.268±.001
Seer + CP-src	.140±.001	.170±.001	.114±.003	.184±.004	.151±.004	.276±.007	.222±.003	.252±.003	.233±.002	.355±.007	.222±.002	.260±.001

Table 11: DISTS (↓) and CLIPIQA (↑, no reference) for the rows of Table 1.

F CHROMA PRESERVATION: CALIBRATION DETAILS

CP-src. $\alpha^\star$ is fitted per checkpoint on 15 fixed clips of the SDSD training split (8 indoor, 7 outdoor, evenly spaced in sorted order; centre 16 frames, 512×512 crop), pooling the second moments of Eq. (7) over all pixels. The average fitted value over the three Seer seeds is $\alpha^\star=(0.20, -0.47, -0.16)$ before clipping.

CP-tgt. Scenes of a target split are sorted by name and split into two folds by index parity (the 10 and 20 fps copies of a BVI-RLV scene and the _2 pass of an SDSD scene stay in the same fold); each clip is fused with the $\alpha^\star$ fitted on the other fold. CP-tgt thus needs paired target data and is not comparable with methods that receive none; we report it separately from Table 1.

Archived coefficient. An earlier version of this work used a single $\alpha=(0.50, 0.50, 0.68)$ for all targets. That value was not produced by Eq. (7) on source data: it was chosen between a closed-form fit on SDSD and one on BVI-RLV by its BVI-RLV test PSNR. We list it for reference only (last row of Table 12); it is not used for any claim.

	L0		L1		L2		L3		L4		L5	
Seer output	PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS
Raw prediction	27.34±0.27	.124±.002	21.93±0.55	.181±.004	17.93±0.11	.251±.002	16.27±0.20	.336±.002	16.07±0.09	.350±.002	18.51±0.21	.343±.0
CP-src ($\alpha^\star$ on SDSD train clips)	27.31±0.26	.126±.002	21.86±0.50	.184±.003	17.97±0.12	.253±.003	16.28±0.20	.325±.006	16.08±0.09	.346±.004	18.49±0.22	.331±.0
CP-tgt ($\alpha^\star$ cross-fitted on target)	27.40±0.14	.134±.008	21.76±0.61	.201±.010	18.78±0.06	.235±.001	16.30±0.20	.327±.002	16.76±0.05	.297±.005	18.57±0.22	.339±.0
CP, archived $\alpha=(.50, .50, .68)$	27.02±0.33	.182±.005	21.73±0.55	.225±.005	18.96±0.09	.242±.002	16.34±0.23	.288±.004	16.62±0.05	.302±.002	18.13±0.20	.310±.0

Table 12: All CP calibrations on the final Seer checkpoints (mean ± s.d. over three seeds). The archived row uses a coefficient selected with BVI-RLV test data (see text) and is shown for reference only.

Is CP specific to Seer? Table 13 applies both calibrations, with the identical protocol and fit clips, to every baseline. For each method CP-src is fitted with the checkpoint the method uses on SDSD, so for URCSA and HVI-CIDNet, whose LOL results come from separate LOL checkpoints, the source-fitted coefficient is transferred across checkpoints; the large drops in those cells reflect that mismatch.

G FULL TEMPORAL METRICS

Method	CP	L0	L1	L2	L3	L4	L5
Zero-DCE	src	**+0.11**	**-0.23**	+0.10	-0.02	**+0.17**	**-0.13**
	tgt	**+0.11**	**+0.08**	**+0.11**	-0.03	**+0.23**	+0.00
SCI	src	-0.05	-0.05	**+0.23**	**-0.07**	**+0.26**	**-0.06**
	tgt	+0.00	+0.04	+0.34	-0.01	**+0.47**	+0.00
HVI-CIDNet	src	+0.06	+0.14	**-0.05**	**-1.95**	**-0.62**	**-1.53**
	tgt	-0.02	+0.18	-0.01	-0.10	+0.00	**+0.20**
DarkIR	src	-0.02	**-0.05**	**-0.05**	**-0.11**	**-0.05**	**-0.15**
	tgt	+0.00	+0.00	+0.00	+0.00	+0.00	**+0.00**
URCSA	src	+0.13	**-0.34**	-0.03	**-0.89**	**+0.17**	**-2.83**
	tgt	-0.14	+0.41	**-0.07**	+0.03	**+0.55**	+0.04
STCD	src	-0.13	+0.10	**-0.15**	-0.04	**+0.16**	**-0.22**
	tgt	-0.14	+0.12	**+0.34**	**+0.15**	**+0.74**	+0.01
Retinexformer	src	**+0.13**	**+0.17**	**+0.05**	**+0.08**	**+0.07**	+0.00
	tgt	+0.04	**+0.76**	-0.01	-0.10	**+0.21**	-0.04
w/o V-JEPA	src	+0.03	**-0.06**	+0.07	+0.03	**+0.03**	**-0.03**
	tgt	+0.10	+0.01	**+0.63**	+0.02	**+0.46**	**+0.05**
Seer	src	-0.03	**-0.07**	**+0.04**	+0.01	+0.02	**-0.02**
	tgt	+0.06	-0.17	**+0.85**	+0.03	**+0.69**	**+0.06**

Table 13: PSNR change (dB) from applying CP to each method's own raw output, with $\alpha^\star$ fitted on SDSD training clips (src) or cross-fitted on the target split (tgt). Bold: 95% bootstrap interval over sequences excludes 0.

	L0 SDSD-in				**L1** SDSD-out				**L2** BVI-RLV			
Method	tLP	WE	MABD	flk	tLP	WE	MABD	flk	tLP	WE	MABD	flk
GT	.030	.0065	.0012	.027	.037	.0095	.0003	.011	.069	.0057	.0023	.089
LQ input	.039	.0062	.0005	.011	.036	.0061	.0003	.006	.061	.0056	.0012	.042
HVI-CIDNet	.180	.0406	.0246	.051	.133	.0331	.0190	.031	.095	.0159	.0094	**.080**
DarkIR	.132	.0183	.0061	.033	.112	.0200	.0042	.018	.081	**.0088**	**.0026**	.086
URCSA†	.118	.0169	.0026	.032	.094	.0172	**.0015**	.016	.084	.0101	.0030	.080
STCD†	.051	.0117	.0024	.031	.067	.0166	.0023	.016	.073	.0115	.0032	.094
Retinexformer†	.095	.0148	.0024	.032	.052	.0140	.0073	.015	.079	.0111	.0029	.093
Retinexformer (retrained)	.090	.0165	.0053	.032	.081	.0201	.0086	.018	.083	.0117	.0029	.093
Pixel branch w/o V-JEPA	.054±.003	.0107±.0003	.0021±.0003	.028±.001	.051±.004	**.0123±.0011**	.0028±.0003	**.012±.001**	.073±.000	.0090±.0003	.0028±.0002	.092±.004
Seer	**.047±.001**	**.0095±.0002**	**.0019±.0000**	**.028±.001**	**.050±.002**	.0131±.0002	.0035±.0002	.013±.000	**.072±.001**	.0089±.0004	.0031±.0003	.092±.001
Seer + CP-src	.052±.004	.0099±.0004	.0019±.0000	.028±.001	.054±.002	.0136±.0004	.0034±.0002	.014±.000	.074±.002	.0094±.0007	.0031±.0003	.092±.001

Table 14: All temporal metrics on the 16-frame clips (↓; flk: per-pixel luma standard deviation over time). tLP, MABD and flk do not account for motion and favour outputs that change little; WE compensates motion with RAFT flow from the ground truth. GT and LQ rows bound the range.